

Mahmoud Alyosify

AI & Data Science Engineer | MSc Artificial Intelligence @ Queen's University, Canada

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🌐 LinkedIn | 🐙 GitHub | 🤖 HuggingFace | 🏆 Kaggle | 🌐 Portfolio

PROFESSIONAL SUMMARY

AI & Data Science Engineer building end-to-end intelligent systems across **LLM fine-tuning**, **self-supervised learning**, **reinforcement learning**, and applied predictive modeling. Currently pursuing an MSc in Artificial Intelligence at **Queen's University, Canada** through the **Digilians Presidential Scholarship**. Developed production-oriented AI projects spanning OSINT threat intelligence on **AWS**, retrieval-ready vision pipelines, deep RL for cloud autoscaling, and time-critical satellite collision risk prediction. Also taught **10,000+ learners** through Udemy and YouTube (**700K+ views**). Interested in LLM systems, MLOps, efficient inference, and scalable AI products with real-world impact.

EDUCATION

MSc in Artificial Intelligence

Queen's University, Canada

Sep 2025 – Present

In Progress

- **Digilians Presidential Scholarship** — competitive national AI graduate fellowship.
- Relevant graduate study in **LLMs, GenAI, Cloud Computing, Deep Learning, Reinforcement Learning**, and **End-to-End ML Pipelines**.

B.Sc. Computer & Information Science (Bioinformatics)

Assiut University

Graduated Jul 2023

GPA: 3.53 / 4.00

- Graduation Project: **Vitalism Solution** — contactless vital sign estimation via rPPG; **Grade: A+; Semi-Finalist, Microsoft Imagine Cup 2023**.

TECHNICAL SKILLS

AI / LLMs

LLMs · Fine-Tuning (LoRA, QLoRA) · RAG · Transformers · Hugging Face · LangChain · Prompt Engineering · Multi-Agent Systems

ML / DL

PyTorch · TensorFlow · Scikit-Learn · XGBoost · LightGBM · CNNs · Computer Vision · NLP · Optuna · SHAP

Data / MLOps

Pandas · NumPy · PySpark · ETL Pipelines · Feature Engineering · Docker · CI/CD · AWS

Backend / Deployment

FastAPI · Streamlit · FAISS · ChromaDB · ONNX · MySQL · MongoDB

Programming

Python · C++ · C · Go · JavaScript · C# · MATLAB · Linux · Git/GitHub

WORK EXPERIENCE

Machine Learning Instructor

National Telecommunication Institute (NTI)

Jul 2025 – Sep 2025

Cairo, Egypt

- Delivered a **90-hour** hands-on ML program covering supervised learning, PCA, and neural networks for engineering and CS students.

AI in Cybersecurity Intern

e& Egypt — via NTI (OJT)

Jun 2025 – Jul 2025

Cairo, Egypt

- Applied AI-driven cybersecurity tools and workflows for threat intelligence, digital forensics, and enterprise security analysis.

Systems Engineer (Military Service)

National Company for Roads Building & Development

Feb 2024 – Mar 2025

Cairo, Egypt

- Supported backend systems for national toll infrastructure serving **50%+ of Egyptian vehicles** and **70%+ of roads**.
- Co-developed an internal desktop application for data management and reporting using **C#, .NET, Entity Framework, SQL Server**, and **Crystal Reports**.

Online Instructor — AI, Algorithms & Data Science

Udemy & YouTube

2020 – Present

- Taught **10,000+ learners** through Arabic technical content; accumulated **700K+ YouTube views**.

SELECTED RESEARCH & ENGINEERING PROJECTS

Horus-OSINT — Cloud-Based Threat Intelligence Assistant

Mar 2026 – Apr 2026 · Python · Llama-3-8B · Fine-Tuning · AWS

- Fine-tuned **Meta-Llama-3-8B** on **159,826** GTD/GDELT records for an OSINT threat intelligence assistant.
- Deployed on **AWS** as an interactive application for querying global terrorism and event intelligence data.

SimCLR-Vision-SSL — Self-Supervised Contrastive Learning

Apr 2026 – Jun 2026 · PyTorch · SimCLR · SupCon · ONNX · FAISS

- Built a self-supervised vision pipeline with a **42-experiment augmentation sweep**, a semi-supervised **SupCon** extension, and deployment-ready **ONNX/FAISS** integration.
- Achieved **84.30% top-1 accuracy**; evaluated representations for classification and retrieval tasks.

Autonomous Cloud Resource Provisioning via Reinforcement Learning

May 2026 – Jun 2026 · Python · PPO · DQN · Gymnasium · Cloud Optimization

- Designed a custom **Gymnasium** environment for autoscaling decisions and trained **PPO** and **DQN** agents for adaptive cloud resource allocation.
- Optimized the trade-off between **infrastructure cost** and **latency** under changing workload conditions.

SCRAP — Satellite Collision Risk Assessment & Prediction

Jan 2026 – Mar 2026 · Python · Scikit-learn · Applied ML · Space Safety

- Built a supervised ML pipeline for early satellite collision risk prediction using conjunction data available **48+ hours** before closest approach, matching time-critical operational decision windows.

Automated Multimodal Agent — PDF to Narrated PowerPoint

Jan 2026 – Mar 2026 · Python · Generative AI · Multimodal Systems · Agentic AI

- Engineered an autonomous multimodal workflow that converts PDFs into narrated PowerPoint presentations by coordinating content extraction, slide generation, and audio narration end-to-end.

Analyzing Review Effort: Human vs. Agentic Pull Requests

Mar 2026 – May 2026 · GitHub Analytics · Data Mining · Empirical Software Engineering

- Built a reproducible analytics pipeline to compare review effort in agent-generated versus human-authored pull requests across large-scale open-source GitHub repositories.

Other Projects

- **AudioShield** — CNN-based deepfake audio detection with Streamlit deployment.
- **Additional:** Vitalism Solution · Diabetes Onset Prediction · E-Commerce Web App · ATM Transaction Analysis · Mini-FAT File System · Student Affairs System.

CERTIFICATIONS

AWS Academy — Data Engineering

AWS Academy — ML Foundations

HCIP — AI (Huawei)

AI in Cybersecurity — NTI (420 hrs)

Full Stack Diploma (.NET & Angular)

AWS Academy — Machine Learning for NLP

Machine Learning Specialization — DeepLearning.AI

Microsoft Certified: Data Analyst Associate

ML Internship — ITIDA / Egypt Makes Electronics (200 hrs)

Generative AI & Prompt Engineering — IBM/Coursera

ACHIEVEMENTS & AWARDS

1st Place — Smart Cities Hackathon 2022

Ideal Student Award — 2022–2023

Semi-Finalist — Microsoft Imagine Cup 2023

1st Place — Science & Tech Content Competition

LEADERSHIP & ACTIVITIES

- Secretary, Higher Scientific & Technological Committee — Student Union, Assiut University.
- Ambassador — Information Technology Institute (ITI), Assiut.
- Volunteer Team Leader — USAID University Center for Career Development.

LANGUAGES

Arabic: Native | **English:** Professional Working Proficiency
